COMPUTER MUSIC CREATION: DEVELOPING WAYS OF THINKING AND NOTATING THE MUSICAL NEEDS, THOUGHTS AND IDEAS OF CREATORS AND PERFORMERS

Andrew Blackburn
Independent Artist
Melbourne

ABSTRACT

What's in a score? Is there more in a score than what the score is itself, more than the score as a noumenon, as something that happens in this world, more than its intrinsic qualities and outer form? Make the invisible, visible? The score invites the artist to negotiate between notions of the map and the terrain. It invites him or her on a journey, following a path, more or less defined, but the landscape and the experience will never be exactly the same due to human and natural dynamics.

A quotation from Kathleen Coessens (2013) is the launching point for a consideration of the need to develop an ontology of notation within the computer music context. Current notational practice within computer music is individualistic, often unique to a piece, and manifest by being built around the specific needs of that composition and perhaps, in a musique-mixte context, the needs of an instrumental performing score. A survey of examples of notation in computer music from the last half century will serve to prove this point.

An exploration of the potential ontological requirements of a semiotically rich notation system argues in favour of the evolution of a language that can be commonly understood. Computer music, as a musical genre, would benefit in its accessibility–to both performers and audiences alike–from such a development, whilst a commonly adopted notation would further improve practitioners and performers ability for dialogue and discourse. Computer language (e.g. ProLogic FORTH or Basic) ontologically have sets of symbols and syntaxes that specify what is to be accomplished, in a form that both computers and their programmers can understand. So, a knowledge of the language provides computer programmers and (literate)users, an agreed vocabulary with which to share ideas, concepts and new forms of knowledge. The language is dynamic as it is continuously evolving to accommodate the changing nature of computing itself.

Using a common set of symbols, perhaps derived from both the symbols of the aforementioned computer programming languages, and from the common music notation syntax could help a computer music language semiotically evolve.

While this paper will not attempt to ascribe or propose a new notation, suggesting a direction practitioners might adopt would be a step in drawing together a vocabulary and conceptual vocabulary for a music computer language.

In musique-mixte, the current practice of recording and sharing a composition is as varied as the contexts in which we find this environment. Just what constitutes a 'score' is perhaps uncertain; - is the score a piece of paper with recognisable symbols, a piece of text, graphics, a software package or patch, or something else entirely? The notation needs of an acoustic musician performing and interacting with computer are often idiosyncratically realised - with many different forms of indication inserted within the score. While the information may be gesturally indicated (e.g. use of a foot pedal to change a parameter or scene change) or through some kind of graphic representation, rarely does the score notation include anything like the procedural information that describes how an effect or process is to be achieved.

There is also a set of electroacoustic music creations which are 'fixed' - that is, the composition is a tape or software that is 'recorded' once and that is then shared. Unlike creations that have a performative or recreative component, as in musique-mixte, notation in this context is largely purposed for analysis. As a tool, software packages such as eAnalyse or SonicVisualiser provide excellent means for the deconstruction and exposition of various layers and structural elements within such a composition. These softwares are essentially historical in their function - they visually analyse sounds that have already occurred - rather than the essential element of a score which is characterised by the affordances that are inherent within it. Within the softwares however, there is already an evolving and partially accepted set of symbols and graphics that may be adaptable.

The development of a semiotic ontology for computer music will enable a language that will permit creators, performers and musicologists have meaningful, precise discourse. It could permit the use of terms which have an agreed definition and, in an age when computer music has moved beyond the experimental and into the more mainstream of music creation and experience, provide a clarity of communication between "experts and aficionados" alike.